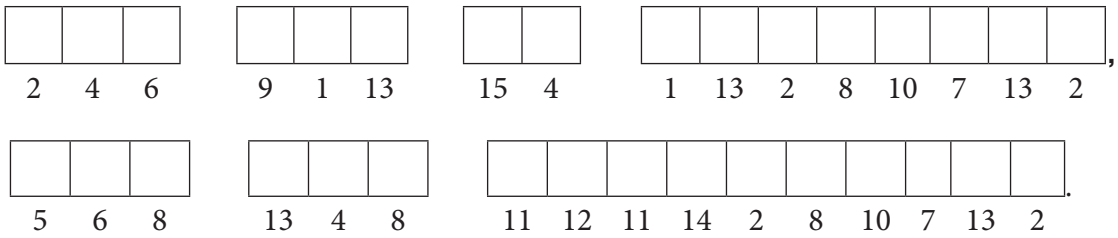
WORKSHEET

Rotations



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Questions

Express the following rotations about the origin as matrices.

1 A rotation of 60°

2 A rotation of $\frac{\pi}{3}$

3 A rotation of $-45°$

4 A rotation of $\frac{5\pi}{3}$

Questions **5** and **6** to two decimal places:

5 A rotation of 140°

6 A rotation of 4 radians

Find the image of each of the following points under the given rotation about the origin, correct to two decimal places.

7 (6, 2) after a rotation of $-\frac{\pi}{6}$

8 (5, 5) after a rotation of 120°

9 $(-3, -4)$ after a rotation of $\frac{3\pi}{4}$

10 (8, 15) after a rotation of 30°

11 $(8, -6)$ after a rotation of $-\frac{\pi}{3}$

12 (7, 3) after a rotation of 74°

Find the image of each of the following geometric shapes after the given rotation.

13 $A(5, 1)$, $B(0, -4)$, $C(-3, 0)$ after a rotation of $\frac{\pi}{2}$.

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14 $A(2, 6), B(4, 1), C(-1, -2)$ after a rotation of 60°

15 $A(5, 5), B(-4, 4), C(0, -3)$ after a rotation of $-\frac{3\pi}{4}$

Answers

A $\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

B $\begin{bmatrix} -0.77 & -0.64 \\ 0.64 & -0.77 \end{bmatrix}$

C $(4.95, 0.71)$

D $A'(0, -5\sqrt{2}), B'(4\sqrt{2}, 0), C'\left(\frac{-3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$

E $(-1.20, -9.93)$

G $\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$

H $(-16.26, -4.95)$

I $(6.20, -1.27)$

N $A'(-1, 5), B'(0, -4), C'(-3, 0)$

O $\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

R $A'(1 - 3\sqrt{3}, 3 + \sqrt{3}), B'\left(\frac{4 - \sqrt{3}}{2}, \frac{1 + 4\sqrt{3}}{2}\right), C'\left(\frac{2\sqrt{3} - 1}{2}, \frac{-2 - \sqrt{3}}{2}\right)$

T $(-6.83, 1.83)$

U $\begin{bmatrix} -0.65 & 0.76 \\ -0.76 & -0.65 \end{bmatrix}$

V $(-0.95, 7.56)$

Y $\begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$

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